

# Chapter 10: Tangible Non-current Assets and Depreciation

Excel file for this chapter: [excel/Ch10\\_Depreciation.xlsx](#). The worked examples below come from its orange tabs; questions 10.9 to 10.13 are self-marking on the blue tabs.

All amounts are in Malaysian Ringgit (RM).

## 1. Learning goals

1. Tell the difference between non-current assets and current assets.
2. Tell the difference between capital expenditure and revenue expenditure.
3. Explain what depreciation is and why it is charged.
4. Calculate depreciation using the straight line, reducing balance and revaluation methods.
5. Record depreciation in the ledger and in the financial statements.
6. Record the sale (disposal) of a non-current asset and find the profit or loss.

## 2. Key words

| Term                       | Meaning                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Non-current asset          | An asset bought to <b>use</b> in the business for <b>more than one year</b> , not to resell. Example: premises, machinery, vehicles. |
| Tangible                   | You can touch it. Example: a van. (Intangible: goodwill, a brand.)                                                                   |
| Cost                       | Purchase price <b>plus</b> all costs to get the asset ready for use: delivery, installation, legal fees, first registration.         |
| Depreciation               | The part of the cost of a non-current asset used up in one year. It is an <b>expense</b> .                                           |
| Residual value             | The amount you expect to sell the asset for at the end of its useful life. Also called scrap value.                                  |
| Useful life                | The number of years the business expects to use the asset.                                                                           |
| Provision for depreciation | Also called accumulated depreciation. The <b>total</b> depreciation charged on an asset since it was bought.                         |
| Carrying amount            | Cost minus accumulated depreciation. Also called net book value (NBV).                                                               |
| Disposal                   | Selling or scrapping a non-current asset.                                                                                            |

### 3. Capital and revenue expenditure

| Capital expenditure                                                                   | Revenue expenditure                                           |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Buying or improving a non-current asset                                               | Day to day running costs                                      |
| Benefit lasts more than one year                                                      | Benefit used up within the year                               |
| Shown in the <b>Statement of Financial Position</b>                                   | Shown as an expense in the <b>Statement of Profit or Loss</b> |
| Examples: buying a van, delivery and installation of a machine, building an extension | Examples: fuel, repairs, insurance, wages, road tax           |

**Why it matters:** if you record capital expenditure as an expense, profit is **understated** and assets are **understated**. If you record revenue expenditure as an asset, profit and assets are **overstated**.

Capital **receipts** (for example, money from selling an old van, or a loan) are not revenue. Revenue **receipts** (sales, rent received, commission) go to the Statement of Profit or Loss.

### 4. Why do assets lose value?

Four causes of depreciation:

1. **Wear and tear:** the asset is used and wears out (a delivery van).
2. **Passage of time:** a lease runs out after a fixed number of years.
3. **Obsolescence:** the asset becomes out of date because newer models exist (old computers).
4. **Depletion:** the asset is used up (a mine or quarry).

### 5. Why charge depreciation?

- **Accruals (matching) concept:** the cost of the asset is spread over the years it helps earn revenue.
- **Prudence:** assets are not overstated in the Statement of Financial Position.
- It gives a true figure for profit each year.

Remember: depreciation is **not** cash. No money leaves the bank when you charge depreciation.

### 6. Method 1: Straight line

The same amount is charged every year.

**Formula:**

$$\text{Annual depreciation} = (\text{Cost} - \text{Residual value}) / \text{Useful life in years}$$

Or a fixed percentage of **cost** each year.

#### Worked example 10.1

A machine costs RM20,000. Expected residual value is RM2,000 after 4 years.

$$\text{Annual depreciation} = (20,000 - 2,000) / 4 = \text{RM4,500 per year}$$

| Year | Depreciation for year (RM) | Accumulated depreciation (RM) | Carrying amount at end (RM) |
|------|----------------------------|-------------------------------|-----------------------------|
| 1    | 4,500                      | 4,500                         | 15,500                      |
| 2    | 4,500                      | 9,000                         | 11,000                      |
| 3    | 4,500                      | 13,500                        | 6,500                       |
| 4    | 4,500                      | 18,000                        | 2,000                       |

**Best for:** assets which give the same benefit each year, for example premises or furniture.

## 7. Method 2: Reducing balance

A fixed percentage of the **carrying amount** (not cost) is charged each year. The charge is high in early years and falls each year.

**Formula:**

$$\text{Depreciation} = \text{Carrying amount at start of year} \times \text{percentage}$$

### Worked example 10.2

Equipment costs RM10,000. Depreciation is 20% per year, reducing balance.

| Year | Carrying amount at start (RM) | Depreciation (RM) | Accumulated depreciation (RM) | Carrying amount at end (RM) |
|------|-------------------------------|-------------------|-------------------------------|-----------------------------|
| 1    | 10,000                        | 2,000             | 2,000                         | 8,000                       |
| 2    | 8,000                         | 1,600             | 3,600                         | 6,400                       |
| 3    | 6,400                         | 1,280             | 4,880                         | 5,120                       |
| 4    | 5,120                         | 1,024             | 5,904                         | 4,096                       |

**Best for:** assets which lose more value in the early years and need more repairs later, for example motor vehicles. High depreciation early plus low repairs balances with low depreciation later plus high repairs.

## 8. Method 3: Revaluation

Used for many small, low cost items such as loose tools. It is too slow to depreciate each item.

**Formula:**

$$\text{Depreciation} = \text{Opening value} + \text{Purchases in the year} - \text{Closing value (valuation)}$$

### Worked example 10.3

|       | Particulars                       | RM | RM | RM         |
|-------|-----------------------------------|----|----|------------|
|       | Loose tools at start of year      |    |    | 1,200      |
| Add:  | Tools bought during the year      |    |    | 500        |
|       |                                   |    |    | 1,700      |
| Less: | Loose tools valued at end of year |    |    | (1,100)    |
|       | <b>Depreciation for the year</b>  |    |    | <b>600</b> |

## 9. Buying part way through the year

Businesses choose a policy and use it consistently:

- **Full year** charge in the year of purchase and none in the year of sale, **or**
- **Monthly (time) basis:** charge only for the months owned.

Example: a machine costing RM12,000 is bought on 1 April. Depreciation is 10% per year on cost. Year end is 31 December.

Full year charge =  $12,000 \times 10\% = \text{RM}1,200$ . Monthly basis for 9 months =  $1,200 \times 9/12 = \text{RM}900$ .

Always read the question to see which policy to use.

## 10. Recording depreciation in the ledger

The asset stays at **cost** in its own account. Depreciation builds up in a separate account.

| Entry                     | Debit                                              | Credit                             |
|---------------------------|----------------------------------------------------|------------------------------------|
| Depreciation for the year | Statement of Profit or Loss (depreciation expense) | Provision for depreciation account |

### Worked example 10.4 (using example 10.1)

Dr      Machinery Account      Cr

| Date         | Particulars | Folio | Amount (RM)   | Date          | Particulars | Folio | Amount (RM)   |
|--------------|-------------|-------|---------------|---------------|-------------|-------|---------------|
| Year 1 Jan 1 | Bank        | CB1   | 20,000        | Year 1 Dec 31 | Balance c/d |       | 20,000        |
|              |             |       | <b>20,000</b> |               |             |       | <b>20,000</b> |
| Year 2 Jan 1 | Balance b/d |       | 20,000        |               |             |       |               |

Dr      Provision for Depreciation of Machinery Account      Cr

| Date          | Particulars | Folio | Amount (RM)  | Date          | Particulars      | Folio | Amount (RM)  |
|---------------|-------------|-------|--------------|---------------|------------------|-------|--------------|
| Year 1 Dec 31 | Balance c/d |       | 4,500        | Year 1 Dec 31 | Income statement | GJ1   | 4,500        |
|               |             |       | <b>4,500</b> |               |                  |       | <b>4,500</b> |
| Year 2 Dec 31 | Balance c/d |       | 9,000        | Year 2 Jan 1  | Balance b/d      |       | 4,500        |
|               |             |       |              | Year 2 Dec 31 | Income statement | GJ2   | 4,500        |
|               |             |       | <b>9,000</b> |               |                  |       | <b>9,000</b> |
|               |             |       |              | Year 3 Jan 1  | Balance b/d      |       | 9,000        |

### Statement of Financial Position (extract) at the end of Year 2

| Particulars               | RM          | RM                              | RM                     |
|---------------------------|-------------|---------------------------------|------------------------|
|                           | <b>Cost</b> | <b>Accumulated Depreciation</b> | <b>Carrying Amount</b> |
| <b>Non-current assets</b> |             |                                 |                        |
| Machinery                 | 20,000      | 9,000                           | 11,000                 |

## 11. Disposal of a non-current asset

When an asset is sold, open a **Disposal account**. It works out the profit or loss on sale.

| Step                         | Debit                      | Credit        | Amount                           |
|------------------------------|----------------------------|---------------|----------------------------------|
| 1. Transfer the cost         | Disposal                   | Asset account | Original cost                    |
| 2. Transfer the depreciation | Provision for depreciation | Disposal      | Accumulated depreciation to date |
| 3. Record money received     | Bank (or Cash)             | Disposal      | Sale proceeds                    |
| 4. Balance the account       |                            |               | Profit or loss                   |

- If the **credit** side is bigger, the difference is a **profit** on disposal (income).
- If the **debit** side is bigger, the difference is a **loss** on disposal (expense).

**Quick check:** Profit or loss = Sale proceeds - carrying amount at date of sale.

### Worked example 10.5

A van cost RM30,000. Accumulated depreciation to the date of sale is RM18,000. It is sold for RM10,000 cash paid into the bank.

Carrying amount = 30,000 - 18,000 = 12,000. Proceeds 10,000. **Loss = 2,000.**

**Dr      Disposal of Van Account      Cr**

| Date         | Particulars | Folio | Amount (RM) | Date         | Particulars                         | Folio | Amount (RM) |
|--------------|-------------|-------|-------------|--------------|-------------------------------------|-------|-------------|
| Date of sale | Van (cost)  | GJ1   | 30,000      | Date of sale | Provision for depreciation          | GJ1   | 18,000      |
|              |             |       |             | Date of sale | Bank                                | CB1   | 10,000      |
|              |             |       |             | Year end     | Income statement (loss on disposal) | GJ2   | 2,000       |
|              |             |       | 30,000      |              |                                     |       | 30,000      |

If the van had been sold for RM13,000, the profit would be RM1,000 and it would appear on the **debit** side as "Income statement (profit on disposal)".

## 12. Where does it all go?

| Item                      | Statement of Profit or Loss          | Statement of Financial Position                                         |
|---------------------------|--------------------------------------|-------------------------------------------------------------------------|
| Depreciation for the year | Expense                              |                                                                         |
| Accumulated depreciation  |                                      | Deducted from cost under non-current assets to give the carrying amount |
| Profit on disposal        | Other income (added to gross profit) |                                                                         |
| Loss on disposal          | Expense                              |                                                                         |
| Cost of asset             |                                      | Non-current assets                                                      |

## 13. Common mistakes

- Using **cost** instead of the **carrying amount** for reducing balance.
- Forgetting to deduct residual value in straight line.
- Recording depreciation in the asset account instead of the provision account.
- Treating repairs as capital expenditure. Repairs keep an asset working; they do not improve it.
- Mixing up the sides of the disposal account. Profit balances on the **debit** side.

## 14. Quick summary

- Non-current assets are used, not sold, and last more than a year.
- Depreciation spreads cost over useful life (accruals) and avoids overstating assets (prudence).
- Straight line: same amount yearly. Reducing balance: percentage of the carrying amount.  
Revaluation: opening + purchases - closing.
- Disposal: proceeds minus carrying amount equals profit or loss.

## 15. Check yourself

1. A computer costs RM3,000 plus RM200 installation. It will last 4 years with no residual value. Find the straight line depreciation.

2. A car costs RM16,000. Depreciation is 25% reducing balance. Find the carrying amount after 2 years.
3. A machine with a carrying amount of RM4,000 is sold for RM4,600. Profit or loss?

Answers: see [model-questions/Answer\\_Key.md](#) , Chapter 10 section.